NOT MEASUREMENT SENSITIVE

MIL-PRF-32359A
23 July 2021
SUPERSEDING
MIL-PRF-32359
23 August 2012

PERFORMANCE SPECIFICATION

CLEANER, GENERAL, FOR GROUND VEHICLES AND GROUND SUPPORT EQUIPMENT, HAP-FREE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers general cleaners (excludes petroleum products) that are hazardous air pollutant (HAP) free and many of which contain either a low content of volatile organic compounds (VOC) or are composed entirely of exempt VOC's, for use on military systems. The word 'cleaners' throughout the specification means both 'general cleaners' and 'cleaning compounds' (more than one component). All classifications and requirements refer to the cleaner at the concentration for which it is to be used.

1.2 Classification. The cleaners are of the following classes, types, and tiers, as specified (see 6.2).

1.2.1 Classes. The classes of cleaners are based on containing or not containing water and are as follows:

Class 1 - Aqueous (contains water).

Class 2 - Non-aqueous (does not contain water).

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM Research Laboratory, Weapons and Materials Research Directorate, Specifications & Standards Office, Attn: FCDD-RLW-MC, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 6850

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1.2.2 Types. The types of cleaners are based on VOC and are as follows:

Type I - VOC exempt/VOC free.

Type II - A VOC level ≤ 25g/L.

Type III - VOC containing.

1.2.3 Tiers. The tiers of cleaners are based on applications types and are as follows:

Tier 1 - Manually sprayed cleaners.

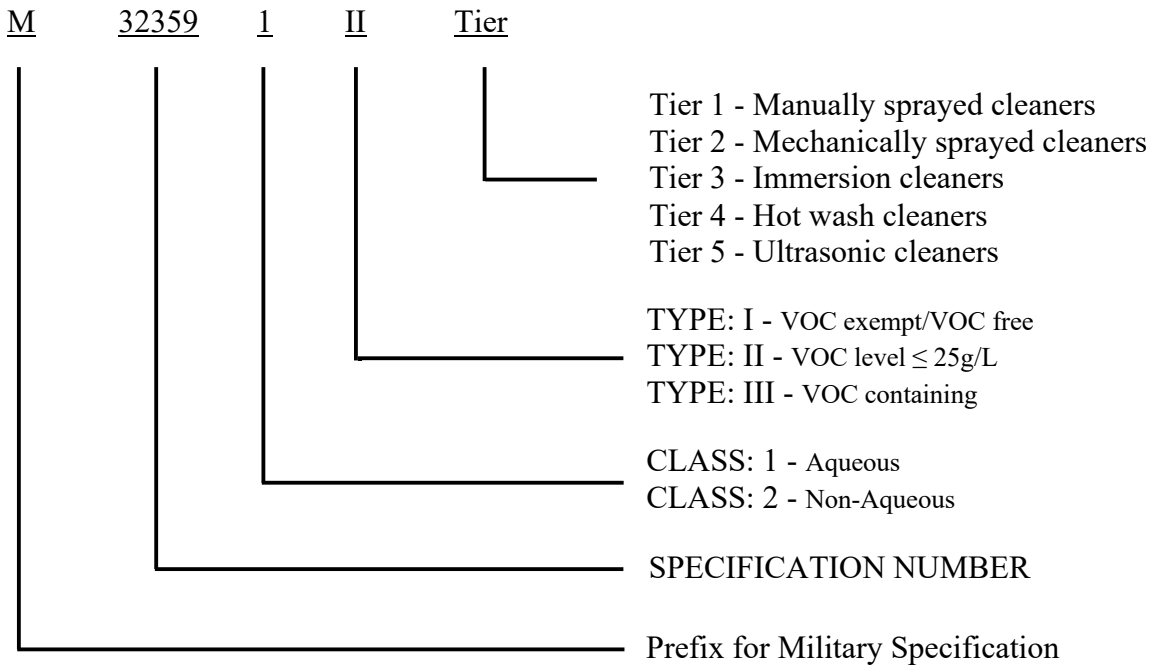
Tier 2 - Mechanically sprayed cleaners, also known as power sprayed.

Tier 3 - Immersion cleaners.

Tier 4 - Hot wash cleaners.

Tier 5 - Ultrasonic cleaners.

1.3 Part or Identifying Number (PIN). PINs to be used for cleaners acquired by this specification are created as follows:



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490	-	Chemical Conversion Coatings and Pretreatments for Ferrous Surfaces (Base for Organic Coatings)

FEDERAL STANDARDS

FED-STD-791	-	Testing Method of Lubricants, Liquid Fuels, and Related Products

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-8625	-	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-PRF-10924	-	Grease, Automotive and Artillery
MIL-DTL-53022	-	Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free
MIL-DTL-53030	-	Primer Coating, Epoxy, Water Based, Lead and Chromate Free
MIL-DTL-53039	-	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-DTL-64159	-	Camouflage Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant

(Copies of these documents are available online at https://quicksearch.dla.mil/ or https://assist.dla.mil)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 63	-	National Emission Standards for Hazardous Air Pollutants for Source Categories
40 CFR 63.750	-	Test methods and procedures.
40 CFR 261	-	Identification and Listing of Hazardous Waste
40 CFR 401.15	-	Toxic pollutants

(Copies of this document are available online at https://www.gpoaccess.gov/cfr/ .)

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SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Method 313	-	Determination of Volatile Organic Compounds (VOC) by Gas Chromatography/Mass Spectrometry (GC/MS)

(Copies of these documents are available online at https://www.aqmd.gov/)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated (see 6.2), the issue in effect on date of invitation for bids or request for proposal should apply.

ASTM INTERNATIONAL

ASTM B209	-	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM E70	-	Standard Test Method for pH of Aqueous Solutions with the Glass Electrode
ASTM F483	-	Standard Practice for Total Immersion Corrosion Test for Aircraft Maintenance Chemicals
ASTM F484	-	Standard Test Method for Stress Crazing of Acrylic Plastics in Contact with Liquid or Semi-liquid Compounds
ASTM F485	-	Standard Practice for Effects of Cleaners on Unpainted Aircraft Surfaces
ASTM F502	-	Standard Test Method for Effects of Cleaning and Chemical Maintenance Materials on Painted Aircraft Surfaces
ASTM F519	-	Standard Test Method for Mechanical Hydrogen Embrittlement Evaluation of Plating/Coating Processes and Service Environments
ASTM F1110	-	Standard Test Method for Sandwich Corrosion Test

(Copies of these documents are available from https://www.astm.org .)

SAE INTERNATIONAL

AMS 2470	-	Anodic Treatment of Aluminum Alloys, Chromic Acid Process
AMS 4377	-	Magnesium Alloy, Sheet and Plate, 3.0Al - 1.0Zn - 0.20Mn (AZ31B-H24), Cold Rolled, Partially Annealed
AMS 4911	-	Titanium Alloy, Sheet, Strip, and Plate, 6Al - 4V, Annealed
AMS 5046	-	Carbon Steel, Sheet, Strip, and Plate, (SAE 1020 and 1025), Annealed
AMS-M-3171	-	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on

(Copies of these documents are available from https://www.sae.org)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict

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between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The cleaning compound furnished under this specification shall be a product which has been tested by an ISO 9000/IEC 17025 or NADCAP certified 3rd party testing facility that is approved by the qualifying activity and has passed the qualification tests specified herein and has been listed or approved for listing on the applicable Qualified Products Database (QPD) by the qualifying activity (see 3.1.1, 3.1.2, 4.2, 4.3, 4.4 and 6.3) before contract award. Any change in the chemical formulation, the material, the process, or the procedure in manufacturing the cleaning compound shall necessitate its being qualified again. The material supplied under contract shall be identical, within manufacturing tolerances, to the products receiving qualification. Any cleaning compound that does not conform to all the qualification tests specified herein shall be removed from the QPD.

3.1.1 Retention of qualification. To retain qualification approval of products listed on the QPD, the manufacturer shall be required to verify by certification to the qualifying activity that its product(s) comply with the requirements of this specification. The time of periodic verification by certification shall be in two-year intervals from the date of original qualification (see 4.3). The certification action is initiated by the qualifying activity.

3.1.2 Qualifying activity. The activity responsible for the collection and storage of cleaning compounds certifications that are required to be furnished under this specification for qualification and conformance testing is the Coatings and Corrosion Team, Materials Development and Transition Branch, Sciences of Extreme Materials Division, ATTN: FCDD-RLW-MC, DEVCOM Army Research Laboratory located at Aberdeen Proving Ground, MD, 21005-5069.

3.2 Material. The materials used in the cleaning compound that are supplied under this specification shall be as specified herein. The materials not specified shall be selected by the contractor and shall be subjected to all the provisions of this specification.

3.2.1 Composition. The composition of the cleaning compound shall be the same as that of the sample furnished for qualification testing. Any change in the formulation of a qualified product shall require requalification. A document specifying the composition shall be forwarded to the Commander, US Army Public Health Command, 5158 Blackhawk Road, Aberdeen Proving Ground, MD 21010-5403.

3.2.2 Prohibited materials. The cleaning compound shall not contain any hazardous ingredients as defined in 40 CFR 261, toxic pollutants as defined in 40 CFR 401.15, nor hazardous air pollutants as defined in 40 CFR 63.

3.2.2.1 Toxicity. The cleaning compound shall not contain any chemical listed by the current report of known carcinogens of the National Toxicology Program (NTP) (see 6.4.1). Any carcinogenic components in the cleaner in a concentration of 0.1 percent or greater by weight or volume will be regarded as sufficient for the presence of a carcinogen in the cleaner. The cleaners supplied under this specification shall have no adverse effect on human health (see 6.5) when used

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for their intended purpose (see 6.1). Army toxicity questions and/or a toxicity clearance request should be addressed to: Commander, US Army Public Health Command.

3.2.2.1.1 Documentation. A certified letter from the U.S. Army Public Health Command, confirming the requirements of this section shall be sent to the Coatings and Corrosion Team, Materials Development and Transition Branch, Sciences of Extreme Materials Division, ATTN: FCDD-RLW-MC, DEVCOM Army Research Laboratory located at Aberdeen Proving Ground, MD, 21005-5069.

3.2.2.2 Hazardous Air Pollutants. The cleaner shall contain no chemicals listed as Hazardous Air Pollutants (HAPs) (see 6.4.2). Any HAP components in the cleaner in a concentration of 1.0 percent or greater by weight or volume will be regarded as sufficient for the presence of HAP in the cleaner. The product containing less than 1% of HAP shall be considered as a HAP-free cleaner.

3.3 Chemical and physical requirements. The cleaning compound shall meet all the requirements listed in table I when tested in accordance with table III.

TABLE I. Chemical and physical requirements.

PROPERTY	TEST METHODS	REQUIREMENTS	
		CLASS 1	CLASS 2
Total Immersion Corrosion Test	ASTM F483	There shall be no indication of staining, etching, pitting, or localized corrosion, or cause weight change greater than that shown in table II (see 4.7.1)	
Sandwich Corrosion Test	ASTM F1110	Rating $\leq$ 1	
Hydrogen Embrittlement	ASTM F519	No failures before 200 hours for Cd coated 4340 steel, Type 1a specimens	N/A
Effect of Cleaners on Unpainted Surfaces	ASTM F485	No streaking, staining not easily removed by hand pressure and water	
Effects on Painted Surfaces	ASTM F502	Coatings to be tested in accordance with table III with no softening in excess of 1 pencil hardness change compared to an equivalent untested, unexposed specimen	
Stress Crazing of Acrylic Plastics, Types A & C	ASTM F484	No crazing or staining	
pH of Aqueous Solutions	ASTM E70	> 3.0 and < 10.0	N/A
VOC content, g/L	SCAQMD Method 313	See type classifications for requirements (see 1.2.2).	
Cleaning Efficiency	4.7.2	The cleaning efficiency shall be greater than 75%.	
Low Temperature Stability	4.7.3	The undiluted cleaning compound shall be homogeneous and shall not separate, layer, or visibly deteriorate.	
Accelerated Storage Stability	4.7.4	No separation or effect on steel strip. Cleaning efficiency shall be no less than freshly made control formula.	

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3.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable (biodegraded) materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

TABLE II. Allowable loss for Total Immersion Corrosion Test.

ALLOYS	ALLOWABLE LOSS (mg/cm^2)
Magnesium - AZ31B-H24, AMS 4377, AMS-M-3171 surface treatment	0.5
Aluminum – 2024-T3, AMS 2470 surface treatment	0.15
Titanium - 6Al-4V, AMS 4911	0.1
Steel - Grade 1020, AMS 5046	0.25
Aluminum - 6061, T6, ASTM B 209	0.2

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).
b. Retention qualification (see 4.3).
c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall consist of all inspections and tests specified in table III. Failure of any test shall be cause for rejection.

4.2.1 Qualification samples. The initial qualification samples shall consist of the required amount of the cleaning compound that is needed by the testing facilities (see 6.3.2). The cleaning compound shall be of the class, type, and tier to be used in filling contract orders. Samples shall be identified (see 6.3.1.1) and forwarded to the appropriate testing facility, and the subsequent testing reports (see 4.2.2) shall be forwarded to the qualifying activity (see 3.1.2) for evaluation.

4.2.2 Test reports. The manufacturer shall provide certified test reports from the agreed upon ISO 9000/IEC 17025 or NADCAP certified laboratory showing methods used and testing results. The material must conform to all the requirements of this specification (see 6.3.1).

4.3 Retention qualification. To ensure compliance with all the requirements of this specification the manufacturer shall complete DD Form 1718 (Certification of Qualified Products) for each plant location and certify that the statements listed on the DD Form 1718 are true. The time of periodic verification by certification shall be in two-year intervals from the date of original qualification (see 3.1.1).

4.4 Conformance inspection. Conformance inspection shall consist of the satisfaction of all of the

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requirements identified as conformance in table III. If a lot fails any of the above acceptance tests during conformance testing, all required qualification tests shall be reinstituted. These qualification tests shall be required until two successive lots meet all requirements of the specification, after which conformance testing shall again be authorized.

4.5 Inspection of materials. The manufacturer is responsible for ensuring that materials and components used are manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification (see 2.4)

4.6 Rejection and retest. When any sample of the product examined and tested in accordance with this specification fails to conform to the requirements specified herein, the entire lot represented by the sample shall be rejected. Rejected material shall not be resubmitted for acceptance without prior approval of the qualifying activity. The application for resubmission shall contain full particulars concerning previous rejections and all measures taken to correct those defects. Samples for retest shall be taken only from a sealed container.

4.7 Test methods. The tests of this specification shall be conducted in accordance with the applicable methods specified in table III and paragraphs 4.7.1 through 4.7.7. The cleaning compound that is required to satisfy all the tests specified in 4.7 shall be selected at random from each lot. The contents of each selected container for sampling shall be thoroughly mixed by rolling and inverting immediately prior to sampling.

4.7.1 Total immersion corrosion. The cleaners shall be tested in accordance with ASTM F 483. The results shall be recorded as an average of three (3) test panels (see table 1).

4.7.2 Soil cleaning test (cleaning efficiency). The test shall be conducted according to the solvent soil test method described in FED-STD-791, Method 7502 (Solvent Cleaning Power by a Soil Test Method), modified by limits on the ultrasonic device. These limits are: The ultrasonic cleaner shall have a frequency of 40 kHz ± 5 kHz and a bath power of 120 watt/gal ± 10 watt/gal. The soil used in this evaluation is MIL-PRF-10924 grease. The acceptable limits for each class of cleaner are defined in table I.

4.7.3 Low temperature stability. Approximately 50 mL of the sample shall be poured into a suitable size test tube and cooled to -18ºC ±5º C (0ºF ± 9°F). This temperature shall be maintained for one hour. The compound shall then be allowed to reach room temperature. After five inversions of the test tube, the compound shall be examined for homogeneity.

4.7.4 Accelerated Storage Stability.

4.7.4.1 Preparation of test sample. Following manufacturer's instructions, a 20.4 oz (600 mL) portion of well-mixed cleaning compound shall be poured into a clean pressure resistant clear glass bottle that is of sufficient size to contain all of the cleaner and to allow insertion of the steel coupon. A strip of steel 6.0 x 0.50 x 0.20 in (15.2 x 1.30 x 0.05 cm) conforming to SAE AMS-5046 shall be polished with 280-grit silicon carbide paper to remove surface contamination and then cleaned by boiling for one minute in mineral spirits and one minute in isopropyl alcohol. The steel strip shall be placed in the test bottle and the bottle capped. The bottle shall then be inverted to completely coat the steel strip.

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TABLE III. Properties and test methods.

PROPERTY	TEST METHOD	REFERENCE PARAGRAPH and/or TABLE	TESTING	
			QUALIFICATION	COMFORMANCE
Total Immersion Corrosion Test	ASTM F483	4.7.1 & Table I	X	X
Sandwich Corrosion Test	ASTM F1110	Table I	X	---
Hydrogen Embrittlement	ASTM F519	Table I	X	---
Effect of Cleaners on Unpainted Aircraft Surfaces	ASTM F485	Table I	X	---
Effects of Cleaning and Chemical Maintenance Materials on Painted Aircraft Surfaces	ASTM F502	Table IV Table I	X	---
Stress Crazing of Acrylic Plastics	ASTM F484	Table I	X	X
Flash Point	ASTM D56	Table I	X	X
pH of Aqueous Solutions	ASTM E70	Table I	X	X
VOC content, g/l	SCAQMD Method 313	Table I	X	X
Cleaning Efficiency	4.7.2 & Table I		X	X
Low Temperature Stability	4.7.3 & Table I		X	---
Accelerated Storage Stability	4.7.4 & Table I		---	---

TABLE IV. Effects on Painted Surfaces ASTM F 502.

COATINGS TO BE TESTED		SURFACE TREATMENT
PRIMER	TOPCOAT	
MIL-DTL-53022	MIL-DTL-64159 or MIL-DTL-53039	MIL-PRF-8625 Type I or Type III
MIL-DTL-53022	MIL-DTL-64159 or MIL-DTL-53039	TT-C-490

4.7.4.2 Procedure. The capped bottle containing the steel strip shall be placed in a hot water bath held at 60°C ±2°C (140°F ±4°F) for 8 hours. The bath and the bottle shall then be allowed to cool for 16 hours and the 24 hour cycle repeated each day for a total of seven days. On the eighth day, the bottle shall be removed from the bath, uncapped, examined for separation, and the steel strip withdrawn and examined for corrosion. The aged sample shall be tested for cleaning efficiency in accordance with 4.7.3.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel components are to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The general cleaners specified by this specification can be used in military ground and support vehicles/equipment, if applicable. In addition, the cleaners specified in this specification were not tested for parts washer cleaning applications. The cleaning method should be identified as either immersion, spray, or hand wipe.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Specify class and type of the cleaner (see 1.2).
c. If issues of documents are different (see 2.2.1, 2.2.2, and 2.3).
d. Specify shelf life requirements (see table I and 6.6).
e. If the size of the conformance sample is different (see 4.5).
f. If conformance tests are to be made on different samples then those specified (see 4.6).
g. If packaging requirements are different (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products Database (QPD)-32359, whether or not such products have actually been so listed by that date. The attention of manufacturers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information (questions, etc.) pertaining to qualification and testing of products and the certified test results from the approved testing facility for the QPD testing (see 4.2) and for the conformance program (see 4.4) should be submitted to the Coatings and Corrosion Team, Materials Development and Transition Branch, Sciences of Extreme Materials Division, ATTN: FCDD-RLW-MC, DEVCOM Army Research Laboratory located at Aberdeen Proving Ground, MD, 21005-5069.

6.3.1 Inspection, and other information. In addition to the qualification test samples, the qualifying activity will request the manufacturer to submit to the qualification activity: (a) a certified test report showing that the material conforms to the requirements of this specification (see 4.2.2); (b) one copy of the SDS (see 6.5); (c) certification that the manufacturer's material meets the HAP

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and VOC content requirement (see table I, 1.2.2 and 3.2.2).

6.3.1.1 Qualification inspection sample identification. Qualification inspection samples are to be forwarded to the laboratory designated in the letter of authorization (see 6.3) and identified as follows:

QUALIFICATION TEST SAMPLES

Specification MIL-PRF-32359 and class 1 or 2; type I, II, or III, tier 1,2,3,4, or 5

CLEANER, GENERAL, FOR GROUND VEHICLES AND GROUND SUPPORT EQUIPMENT, HAP-FREE

Manufacturer's name and product number
Submitted by (name and date) for qualification testing in accordance with authorization (reference authorizing letter)

6.3.2 Testing facilities. The testing facilities identified below can each do some of the required tests, however, all the tests can be performed by a combination of those facilities listed below. Please note that this list is not an exclusive list. The list was prepared for your convenience only. There are other facilities that are capable of performing the required tests. The South Coast Air Quality Management District (SCAQMD) can perform only the VOC Content test.

SMI, Inc.
12219 SW 131 Avenue
Miami, FL 33186-6401 USA
Ph: 305-971-7047
Attn: Patricia D. Viani, President

Petro-Lube Test Labs.
116 Sunset Inn Road
P.O. Box 300 Lafayette, NJ 07848
Ph: 973-579-3448
Attn: Josiah Wintermute

South Coast Air Quality Management District
21865 Copley Dr.
Diamond Bar, CA 91765
Ph: 909-396-2000

6.3.2.1 Test methods' estimated quantity. The quantity and costs for the specific tests required by this document are given below in table V. The approximate time to complete all tests is ten weeks.

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TABLE V. Test Method and Quantity for required tests. 1/

SPECIFIC TEST	TEST METHOD	QUANTITY (mL)
Total Immersion Corrosion Test	ASTM F483 (5 specimens)	1000
Sandwich Corrosion Test	ASTM F1110	200
Hydrogen Embrittlement	ASTM F519 (type 1a specimens)	400
Effect of Cleaners on Unpainted Surfaces	ASTM F485	800
Effects on Painted Surfaces	ASTM F502 (6 specimens)	400
Stress Crazing of Acrylic Plastics	ASTM F484 (types A and C)	200
pH of Aqueous Solutions	ASTM E70	100
VOC content, g/L	SCAQMD Method 313	200
Cleaning Efficiency	FED-STD-791, Method 7502 (MIL-PRF-680, App A)	1200
Low Temperature Stability	MIL-PRF-85570D, section 4.5.8	100
Accelerated Storage Stability	MIL-PRF-85570D, section 4.5.2	1200

1/ A minimum of 6300 ml of cleaner will be required for qualification testing

6.4 Definitions.

6.4.1 Carcinogens. The Occupational Safety and Health Administration (OSHA) definition of carcinogens is those chemicals / processes appearing in lists 1, 2A, and 2B of the International Agency for Research on Cancer (IARC) 1/; substances known to be carcinogenic and occupational exposures associated with a technological process known to be carcinogenic by the National Toxicology Program (NTP) Report on Carcinogens (latest annual report)2/; and OSHA regulated carcinogens3/.

1/ Copies of these volumes may be found in medical libraries or through the World Health Organization, 1211 Geneva 27, Switzerland.

2/ Copies may be obtained from the Public Health Service, National Toxicology Program, Public Information Office, P.O. Box 12233, MD B2-04, Research Triangle Park, NC 27709 or at https://ntp.niehs.nih.gov.

3/ Copies may be found in 29 CFR part 1910 Subpart Z.

6.4.2 Hazardous Air Pollutant (HAP). HAP is defined as any substance listed under Section 112 of the Clean Air Act or its modifications. The text of the Clean Air Act, listed pollutants and modifications are kept by the Environmental Protection Agency (EPA) and are accessible through the website: https://www.epa.gov.

6.4.3 Volatile organic compound (VOC). VOC is defined in 40 CFR Part 51.100 (s). Volatile organic compounds (VOC) means any compound of carbon, excluding carbon monoxide, carbon dioxide, carbonic acid, metallic carbides or carbonates, and ammonium carbonate, which participates in atmospheric photochemical reactions. For the latest complete definition of VOC

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see: https://www.epa.gov/ttn/naaqs/ozone/ozonetech/def_voc.htm.

6.4.3.1 Exempt VOC. A list of exempt VOC is listed in 40 CFR Part 51.100 (s). For the current list of exempt organic compounds see:

https://www.epa.gov/ttn/naaqs/ozone/ozonetech/def_voc.htm.

Compounds that are explicitly exempted from regulation by the EPA as VOCs are those compounds of carbon that have negligible photochemical reactivity (they do not produce ozone in atmospheric reactions).

6.4.4 Hand-wipe cleaning. Hand-wipe cleaning operation means the removal of contaminants such as dirt, grease, oil, and coatings from a substrate by physically rubbing with a material such as a rag, paper, or cotton swab that has been moistened with a cleaning solvent. Refer to MIL-PRF-32405 for a list of requirements.

6.5 Safety Data Sheet (SDS). A SDS must be prepared and submitted in accordance with FED-STD-313. The SDS must also meet the requirements of 29 CFR 1910.1200. The 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Questions pertinent to the effect(s) of these cleaners on the health of personnel using them should be referred by the procuring activity to the appropriate medical service, who will act as its adviser. Contracting officers will identify the activities requiring copies of the SDS.

6.6 Subject term (key word) listing.

Aqueous
Environmental
Non-aqueous
Non-aviation
Non-hazardous

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MR
Navy - SH
Air Force - 68

Review activities:
Army - AT, EA, GL
Navy - AS, CG, MC
Air Force - 03, 50
DLA - GS, GS3

Civil agency:
GSA - FAS
MISC - MP

Preparing activity:
ARMY – MR

(Project 6850-2019-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:53Z
Check the source to verify that this is the current version before use.